

Directional derivative and Gradient vector:-

In this section, we introduce a type of derivative called directional derivative, that enables us to find the rate of change of a function of two or more variables in any direction.

Directional Derivative:-

The directional derivative of f at (x_0, y_0) in the direction of a unit vector

$u = \langle a, b \rangle$ is calculated as:

$$D_u f(x, y) = \nabla f(x, y) \cdot u \quad \nabla f \rightarrow \text{vector fn.}$$

This expresses the directional derivative in the direction of unit vector, u .

where $\nabla f(x, y)$ is a gradient of function of two variables x and y , and defined by,

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle =$$

$$\frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j \quad \text{and}$$

Unit vector u in the direction of v is

$$u = \frac{v}{|v|}$$

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Exp:- Find the directional derivative of the fn:
 $f(x, y) = x^2 y^3 - 4y$ at the point $(2, -1)$ in
 the direction of the vector $v = 2i + 5j$.

Solution:-

We know that directional derivative is
 defined as: $\boxed{D_{\vec{u}} f(x, y) = \nabla f(x, y) \cdot \vec{u}} \rightarrow \text{eq (i)}$

first we find gradient vector $\nabla f(x, y)$ at $(2, -1)$

$$\nabla f(x, y) = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j, \text{ when } f(x, y) = x^2 y^3 - 4y$$

$$\nabla f(x, y) = (2xy^3)i + (3x^2y^2 - 4)j \text{ at } (2, -1)$$

$$\nabla f(x, y) = 2(2)(-1)^3 i + 3(2)^2(-1)^2 - 4j$$

$$\nabla f(x, y) = -4i + 12 - 4j$$

$$\boxed{\nabla f(2, -1) = -4i + 8j} \rightarrow \text{Gradient vector eq (ii)}$$

Now, compute the unit vector in the direction
 of Vector v is,

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{2i + 5j}{\sqrt{(2)^2 + (5)^2}} = \left[\frac{2i}{\sqrt{29}} + \frac{5j}{\sqrt{29}} \right] \rightarrow \text{eq (iii)}$$

Put eq (2) and eq (3) in eq (1)

$$D_{\vec{u}} f(x, y) = (-4i + 8j) \cdot \left(\frac{2}{\sqrt{29}} i + \frac{5}{\sqrt{29}} j \right)$$

$$= \left(\frac{-4i \cdot 2i}{\sqrt{29}} \right) + \left(\frac{8j \cdot 5j}{\sqrt{29}} \right) \quad \because i \cdot i = 1$$

$$= \frac{-8}{\sqrt{29}} + \frac{40}{\sqrt{29}} = \frac{-8 + 40}{\sqrt{29}} = \frac{32}{\sqrt{29}} \quad \begin{matrix} j \cdot j = 1 \\ i \cdot j = 0 \end{matrix}$$

$$\boxed{D_{\vec{u}} f(x, y) = \frac{32}{\sqrt{29}}} \rightarrow \text{Directional derivative of } f(x, y) \text{ at } (2, -1)$$